

Questions Unit 2

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Questions on Stochastic Processes

Conceptual Questions

1. Define a stochastic process and explain the difference between discrete-time and continuous-time stochastic processes.
2. What is the state space of a stochastic process, and how can it differ between different types of processes?
3. Describe the Markov property. How does it simplify the analysis of stochastic processes?
4. Differentiate between a Markov process, a Poisson process, and Brownian motion.
5. What is a stationary process? Give an example where this type of process would be useful.
6. Define a Poisson process and explain its key properties, including the concept of exponential interarrival times.
7. Explain the Chapman-Kolmogorov equations and their significance in Markov chains.
8. What is the difference between recurrent and transient states in a Markov chain? Provide an example of each.
9. Discuss the concept of an absorbing state in a Markov chain and its importance in stochastic modeling.
10. What are the key characteristics of ergodic Markov chains?

Application-Based Questions

1. Describe how you would use a Poisson process to model customer arrivals at a coffee shop. Include assumptions and calculations of probabilities.
2. Given a Markov chain with two states, "Sunny" and "Rainy," and transition probabilities of 70% for staying in the same weather and 30% for changing, compute the probability of three sunny days in a row.
3. In a M/M/1 queue, with arrival rate $\lambda = 3$ customers per hour and service rate $\mu = 5$ customers per hour, compute the steady-state probabilities of having 0, 1, 2, and 3 customers in the system.
4. Calculate the probability of exactly 2 events occurring in the next hour in a Poisson process with an average rate of 5 events per hour.
5. Consider a random walk on a number line where each step is equally likely to move left or right. After 10 steps, what is the probability of being back at the starting point?

Problem-Solving Questions

1. Using the forward algorithm in a Hidden Markov Model, compute the probability of an observation sequence $O = \{3, 1, 2\}$ given the transition and emission probabilities.
2. Solve for the stationary distribution of the following Markov chain with transition matrix:

$$P = \begin{pmatrix} 0 & .80 & .20 \\ .60 & .4 & \end{pmatrix}$$

3. For an M/M/1 queue with $\lambda = 4$ and $\mu = 6$, compute the average number of customers in the system and the average waiting time.
4. Given a birth and death process with constant birth rate $\lambda = 2$ and death rate $\mu = 3$, find the steady-state probabilities for having 0, 1, or 2 individuals in the system.
5. In a CTMC, transitions between states occur with the rate matrix Q . Derive the stationary distribution for a process with the following rate matrix:

$$Q = \begin{pmatrix} - & & \\ & 3 & 3 \\ & 1 & -1 \end{pmatrix}$$

True/False Questions

1. In a Poisson process, the interarrival times are exponentially distributed.
2. A Markov process is memoryless, meaning the future state depends only on the current state and not on the past states.

3. In an M/M/1 queue, the average number of customers in the system is directly proportional to the arrival rate λ .
4. The sum of the probabilities in a transition matrix must equal 1 for each row.
5. A stationary distribution in a Markov chain must satisfy $\pi P = \pi$, where π is the stationary probability vector.

Advanced Conceptual Questions

1. What is the Kolmogorov forward equation (or Fokker-Planck equation) for continuous-time Markov processes, and how is it used to describe the time evolution of the probability distribution?
2. Define and differentiate between discrete-state continuous-time Markov processes and continuous-state continuous-time Markov processes.
3. Explain the concept of a martingale in the context of stochastic processes. Provide an example of a martingale process.
4. What is the generator matrix of a continuous-time Markov chain, and how does it relate to the transition probabilities of the process?
5. Explain the concept of hitting time or first-passage time in Markov processes. How would you calculate it in simple random walks?
6. What are jump processes? Provide an example of a jump process and describe how it differs from continuous processes like Brownian motion.
7. How is the Strong Law of Large Numbers related to stochastic processes, particularly in the context of random walks or queueing models?
8. Define the ergodicity of a Markov chain. Under what conditions is a Markov chain said to be ergodic?

Scenario-Based Questions

1. A CTMC (Continuous-Time Markov Chain) has two states, with transition rates from state 1 to state 2 being 3 and from state 2 to state 1 being 1. Find the stationary distribution and the mean recurrence time of each state.
2. Consider a birth-death process where the birth rates $\lambda_n = 5$ for all n , and the death rates $\mu_n = n$. Calculate the steady-state probabilities and determine the long-run proportion of time the process spends in each state.
3. A Brownian motion process starts at $X_0 = 0$. What is the probability that the process crosses $X = 1$ for the first time before $t = 2$?

4. A Hidden Markov Model (HMM) has two hidden states (Rainy and Sunny) and two observable states (Walk and Shop). If the initial probabilities and the transition and emission probabilities are given, use the Viterbi algorithm to determine the most likely sequence of hidden states for the observation sequence Walk, Shop, Walk.
5. In an M/M/1 queue, if the arrival rate $\lambda = 4$ and the service rate $\mu = 6$, calculate the probability that there are more than 2 customers in the system at steady state.

Calculation-Based Questions

1. A Markov chain has three states with the following transition matrix:

$$P = \begin{pmatrix} 0 & .50 & .30 \\ .20 & .40 & .40 \\ .20 & .20 & .50 \end{pmatrix}$$

Determine the stationary distribution for the system.

2. For a Poisson process with a rate $\lambda = 3$, calculate the probability that no event occurs in the interval $[0, 2]$, and calculate the expected number of events in the interval $[0, 5]$.
3. Given a CTMC with states 0, 1, and 2, and transition rate matrix:

$$Q = \begin{pmatrix} -4 & 4 & 0 \\ 2 & -3 & 1 \\ 0 & 1 & -1 \end{pmatrix}$$

Find the mean hitting time to state 0 starting from state 2.

4. For a queueing system modeled by an M/G/1 queue with arrival rate $\lambda = 4$ and service times that are normally distributed with mean 2 and variance 1, compute the mean waiting time in the system.
5. In a random walk on a 2D grid, starting from the origin, what is the probability that the walk returns to the origin after exactly 4 steps?

Proof-Based Questions

1. Prove that in a discrete-time Markov chain, the stationary distribution π satisfies $\pi P = \pi$, where P is the transition probability matrix and π is the stationary distribution vector.
2. Show that the Poisson process has independent and stationary increments, and derive the distribution of interarrival times for the process.
3. Prove that for a birth-death process, the ratio of the steady-state probabilities $\frac{p_{n+1}}{p_n}$ is equal to $\frac{\lambda_n}{\mu_{n+1}}$, where λ_n is the birth rate and μ_{n+1} is the death rate.

4. Demonstrate that a Brownian motion process is both Gaussian and has independent increments.
5. Prove that a renewal process with exponentially distributed interarrival times is a Poisson process.

True/False Questions

1. In a Markov process, the probability of moving to a future state depends on the entire past history of the system.
2. A Poisson process is always a continuous-time process with exponentially distributed interarrival times.
3. The expected waiting time in an M/M/1 queue is proportional to the service rate.
4. The Viterbi algorithm finds the most likely sequence of hidden states in a Hidden Markov Model.
5. The Kolmogorov backward equations describe the evolution of the transition probabilities in continuous-time Markov chains.